

YOAN KESRAOUI

MACHINE LEARNING AND QUANTITATIVE RESEARCH

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RESEARCH PROFILE

Quantitative researcher trained in data science, econometrics and applied mathematics, with experience applying machine learning to forecasting, designing rigorous evaluation protocols and analysing large-scale transaction data. Seeking to pursue doctoral research in machine learning and computational statistics at Meta FAIR.

EDUCATION

Université Paris 1 Panthéon-Sorbonne - M.Sc. Data Science 2025-2026

Machine learning, deep learning, statistics, optimisation and time series

Université Paris Cité - M.Sc. Applied Economics - Econometrics 2023-2025

Econometrics, statistical inference, financial modelling and panel data

Université Paris Cité - B.Sc. Economics & Mathematics 2020-2023

RESEARCH AND PROFESSIONAL EXPERIENCE

ENGIE Global Markets - Quant Research Intern - Energy Markets Jun 2026-Present

Global Market Analysis, Paris

- Developed a **probabilistic day-ahead forecasting framework for five European power markets**, combining Chronos-2, residual models and adaptive Kalman filtering.
- Fine-tuned **Chronos-2 using LoRA (ranks 8 and 16)** across four markets; trained a weather-conditioned adapter using archived NOAA GFS forecasts.
- Implemented and benchmarked a **sparse mixture-of-experts residual model in PyTorch** to study regime-dependent forecast corrections.
- Implemented 365-day rolling evaluations and time-ordered out-of-fold calibration with fold-specific LoRA refits; tracked data provenance, publication cutoffs and probabilistic metrics.

Société Générale - Quant Market Risk Intern Sep 2025-Mar 2026

Front Office Market Risk, Paris

- Developed **GARCH, EGARCH and VAR** models to study financial time series and portfolio-risk dynamics.
- Built **Python/SQL/API** pipelines for large risk datasets; applied clustering and k-nearest neighbours to anomaly detection and risk monitoring.

Autorité des marchés financiers (AMF) - Data Scientist / Economist Jan-Jun 2025

Financial Stability & Risk Studies, Paris

- Analysed **millions of MiFID II transactions** using PySpark, Python, R and panel econometrics to study investor behaviour and market dynamics.
- Engineered investor-level turnover, return and trading-intensity indicators; contributed empirical analysis to the *AMF Market and Risk Mapping 2025*.

Generali - Quantitative Analyst Intern Apr-Sep 2024

ALM & Strategic Asset Allocation, Paris

- Built **XGBoost surrogate models** for computationally intensive full-revaluation pricing; investigated non-linear relationships between risk factors.
- Calibrated Vasicek/CIR interest-rate models and automated analytical workflows with Python, SQL and Power BI.

TECHNICAL SKILLS

Programming and data: Python, SQL, PySpark/Spark, R, Git

ML libraries: PyTorch, TensorFlow, scikit-learn, CatBoost, XGBoost, pandas, NumPy, statsmodels

Methods: Time-series forecasting, deep learning, econometrics, statistical inference, probabilistic modelling, optimisation, anomaly detection

Languages: French (native), English (C1), Spanish (beginner)